Journal of Youth and Adolescence (2026) 55:2096–2112
https://doi.org/10.1007/s10964-026-02371-4

EMPIRICAL RESEARCH

How Does Mothers' Parental Burnout Longitudinally Predict Adolescent Bullying Perpetration and Victimization? The Mediating Role of Psychological Control Rather than Autonomy Support

Haotian Bian[1] · Qian Pu[1] · Ying He[1] · Yifan Zhang[2] · Fumei Chen[1]

Received: 27 March 2026 / Accepted: 28 May 2026 / Published online: 8 June 2026
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2026

Abstract

Bullying perpetration and victimization are prevalent adolescent peer problems with lasting adverse consequences. Although previous research identifies various parental and family risks for these peer problems, the role of mothers' parental burnout and the underlying longitudinal mechanisms remain underexplored. This three-wave longitudinal study (6-month intervals across 12 months) examined whether higher levels of mothers' parental burnout at T1 predicted higher adolescent bullying perpetration and victimization at T3, and whether adolescents' perceptions of higher psychological control at T2 and lower autonomy support at T2 mediated these associations. The sample included 1,969 adolescents (49.90% boys; M_{age} = 14.35, SD_{age} = 1.30) and their mothers (M_{age} = 41.68, SD_{age} = 5.24). After controlling for autoregressive effects and covariates (gender, school, and SES), results showed that higher mothers' parental burnout at T1 predicted higher psychological control at T2 and lower autonomy support at T2, as well as greater bullying perpetration and victimization at T3. Higher psychological control at T2, rather than lower autonomy support at T2, uniquely predicted greater bullying perpetration and victimization at T3 and emerged as the specific significant longitudinal mediator. The indirect effect through psychological control exceeded that through autonomy support for both outcomes (significant at the 90% CI for bullying perpetration and 95% CI for bullying victimization). These findings revealed that psychological control is the specific parenting mechanism through which mothers' parental burnout contributes to adolescents' later bullying perpetration and victimization.

Keywords Parental burnout · Bullying perpetration · Bullying victimization · Psychological control · Autonomy support · Adolescents

Introduction

Bullying perpetration and victimization remain critical global issues, with prevalence rates among children and adolescents of 16% and 25% (Ariani et al., 2025). Both experiences are associated with severe consequences that persist across the lifespan (Arseneault, 2018; O'Brennan et al., 2009). Although bullying unfolds in peer contexts, adolescents' peer behaviors are also shaped by upstream family processes (Swearer & Hymel, 2015). Parental burnout may represent a significant parental and family risk factor (Mikolajczak & Roskam, 2018). Examining its association with adolescent bullying perpetration and victimization may clarify whether the effects of parental burnout extend beyond intra-familial dynamics to extra-familial peer outcomes. In addition, mothers carry a greater share of daily

✉ Fumei Chen
chenfumei@bnu.edu.cn

Haotian Bian
htbian@mail.bnu.edu.cn

Qian Pu
puqian1228@163.com

Ying He
He_Yinggg@outlook.com

Yifan Zhang
202431061107@mail.bnu.edu.cn

[1] Collaborative Innovation Center of Assessment for Basic Education Quality, Beijing Normal University, Beijing, China

[2] State Key Laboratory of Cognitive Neuroscience and Learning, Beijing Normal University, Beijing, China

caregiving, which heightens their vulnerability to parental burnout and may render its consequences more serious for adolescent developmental outcomes (Wang et al., 2023). However, few longitudinal studies have examined whether mothers' parental burnout predicts adolescent bullying perpetration and victimization, and even fewer have integrated negative and positive parenting behaviors as simultaneous mediators within a longitudinal framework of this family-to-peer process. To address these gaps, this study used three waves of longitudinal data collected at 6-month intervals to examine whether mothers' parental burnout at baseline (T1) predicted adolescent bullying perpetration and victimization at the 12-month follow-up (T3). Adolescents' perceptions of psychological control and autonomy support at the 6-month follow-up (T2) were further examined as potential mediators of these associations.

Theoretical Framework

The parenting process model (Belsky, 1984; Belsky & Jaffee, 2006) and the dual-process model (Vansteenkiste & Ryan, 2013) together provide a useful framework for understanding how multiple determinants shape parenting behavior and how parenting behavior further relates to adolescent development. The parenting process model identifies three major determinants of parenting behavior, including parents' ontogenic origins and personal resources, children's characteristics, and contextual sources of stress and support (Belsky, 1984; Belsky & Jaffee, 2006). Furthermore, parent, child, and social-context characteristics do not exert equal influence. Parents' ontogenic origins and personal resources occupy a primary position. These factors exert a direct influence on parenting and an indirect effect by shaping child characteristics and contextual factors (Belsky & Jaffee, 2006). Parental burnout occurs when parenting resources become chronically depleted and are no longer sufficient to meet the ongoing parenting demands (Mikolajczak & Roskam, 2018). Accordingly, parental burnout may increase reliance on less resource-demanding but maladaptive strategies, such as psychological control, while limiting parents' capacity to provide more resource-intensive positive practices, such as autonomy support.

Furthermore, the dual-process model (Vansteenkiste & Ryan, 2013) distinguishes psychological control and autonomy support as two contrasting forms of parenting. Psychological control undermines adolescents' autonomy development, whereas autonomy support encourages independent behavior (Zheng & Chen, 2025). These two forms of parenting show differential associations with adolescent developmental outcomes (Bradshaw et al., 2025). Specifically, higher psychological control is more strongly associated with greater adolescent maladaptive outcomes, whereas lower autonomy support is more strongly associated with poorer adolescent adaptive outcomes. Cross-domain paths also emerge, with higher psychological control predicting lower adaptive outcomes and lower autonomy support predicting greater maladaptive outcomes, though these associations remain comparatively weaker. Integrating these frameworks, parenting behaviors (e.g., psychological control and autonomy support) may serve as mediating pathways linking mothers' parental burnout to adolescent bullying perpetration and victimization.

Parental Burnout, Bullying Perpetration and Victimization

Bullying perpetration refers to repeated intentional aggression in peer contexts, whereas bullying victimization involves being repeatedly targeted by peers (Olweus, 1993). According to the social-ecological model of bullying (Swearer & Hymel, 2015), both forms are shaped by parental and family risk factors. Insufficient parental and family resources may constitute an important proximal risk factor. Parental burnout refers to an intense sense of exhaustion experienced by parents who are chronically exposed to parenting stress (Roskam et al., 2017). It arises when parenting demands persistently exceed parents' available psychological, social, or material resources (Mikolajczak & Roskam, 2018). Parental burnout occurs across diverse cultural contexts, with a 42-country study estimating its average prevalence at 3.2% (Roskam et al., 2021). Although the prevalence of mothers' parental burnout has not been specifically documented, previous research involving Chinese and German parents found higher levels of parental burnout among mothers than fathers (Teuber et al., 2024). This difference likely reflects gender inequalities in parenting responsibilities, with mothers being more involved in daily caregiving (Roskam et al., 2022). Crucially, the consequences of mothers' parental burnout extend beyond the individual, spilling over into parent-adolescent interactions and potentially impairing adolescents' social adaptation in broader contexts (Bian et al., 2025; Mikolajczak et al., 2018; Yang et al., 2021). When mothers are burned out, their diminished capacity for emotional regulation and supportive parenting can leave adolescents vulnerable to maladaptive peer dynamics. Therefore, examining mothers' parental burnout may help clarify whether parental resource depletion extends to extra-familial peer dynamics, shaping later social adaptation difficulties (e.g., bullying perpetration and victimization).

Despite this theoretical basis, empirical evidence linking mothers' parental burnout directly to adolescent bullying perpetration or victimization in peer contexts remains scarce. Regarding bullying perpetration, studies of broader

adolescent behavioral problems provide relevant evidence. Cross-sectional research has found a positive association between mothers' parental burnout and adolescents' antisocial behavior, whereas fathers' parental burnout showed no such association (Cheng et al., 2020). A longitudinal study of mothers found that parental burnout predicted increases in adolescents' externalizing problems (Chen et al., 2022). Two processes may explain why higher levels of mothers' parental burnout are associated with greater adolescent bullying perpetration. First, burned-out parents tend to withdraw emotional investment and provide primarily instrumental support, potentially undermining adolescents' relational need satisfaction (Bian et al., 2026b). According to the frustration-aggression hypothesis (Berkowitz, 1989; Dollard et al., 1939), unmet psychological needs may lead to frustration, which can be redirected toward peers in the form of bullying perpetration. Second, evidence suggests that parental burnout contributes to more parental hostility and violence (Chen et al., 2022; Mikolajczak et al., 2018), which adolescents may internalize and reproduce in peer interactions, including bullying others, through social learning processes (Bandura, 1977).

Regarding bullying victimization, existing studies also provide relevant evidence. A large-scale cross-sectional survey of fathers and mothers in the United States showed that parenting stress was positively associated with bullying victimization (Zhang et al., 2021). Given that parental burnout reflects a more chronic and overwhelming experience of parenting stress, mothers' parental burnout may be similarly or even more strongly associated with bullying victimization. Furthermore, among fathers and mothers, parental burnout may contribute to neglectful parenting, disrupted parent-child relationships, and reduced family cohesion (Bian et al., 2025; Mikolajczak et al., 2018). Prior studies have further linked these factors to a greater risk of bullying victimization (Fredrick et al., 2022; Nocentini et al., 2019; Zhou et al., 2023). One plausible explanation is that adverse family dynamics associated with parental burnout may undermine adolescents' attachment security and self-esteem (Sciarrino et al., 2018; Song et al., 2024; Zhang et al., 2023). Peers may perceive these characteristics as signs of vulnerability, thereby increasing adolescents' risk of being targeted by bullies (Bian et al., 2026a; Ward et al., 2018). Finally, bullying perpetration and victimization often co-occur and exhibit reciprocal associations over time (Choi & Park, 2021), suggesting that regardless of whether parental burnout initially contributes to bullying perpetration or victimization, the two forms are likely to reinforce each other.

Dual-Process: The Mediating Roles of Psychological Control and Autonomy Support

According to the parenting process model and the dual-process model (Belsky, 1984; Belsky & Jaffee, 2006; Vansteenkiste & Ryan, 2013), psychological control and autonomy support may serve as potential mediators linking mothers' parental burnout to adolescent developmental outcomes. Several studies have examined these links. For instance, a time-lagged study spanning one school semester showed that mothers' parental burnout predicted higher psychological control, which in turn predicted greater social avoidance and distress among adolescents (Wang et al., 2023). Similarly, mothers' parental burnout at baseline predicted lower autonomy support 6 weeks later, which subsequently predicted more social adjustment problems at 12-week follow-up (Song et al., 2024). However, longitudinal research still needs to clarify how psychological control and autonomy support may link mothers' parental burnout to extra-familial bullying perpetration and victimization.

Existing longitudinal studies have shown two patterns: mothers' parental burnout predicts increases in psychological control and decreases in autonomy support over time (Song et al., 2024; Wang et al., 2023). These two patterns can be understood through the perspective of the parenting process model (Belsky, 1984; Belsky & Jaffee, 2006). Psychological control may offer burned-out parents a low-cost strategy for maintaining adolescent compliance with minimal emotional involvement, whereas autonomy support is highly resource-demanding. Specifically, autonomy support requires sufficient psychological, social, and material resources, including sustained perspective-taking, patience, time, practical support, and unconditional acceptance; therefore, burned-out parents may struggle to consistently provide it. Taken together, when burned-out parents seek to balance parenting demands with depleted parenting resources, they may resort to psychological control while reducing autonomy-supportive parenting.

On the other hand, broader longitudinal studies have shown that higher psychological control predicts increases in bullying perpetration and victimization over time (Peng et al., 2024; Ren et al., 2023; Zhang et al., 2025), while lower autonomy support predicts higher bullying perpetration and victimization (Rajendran et al., 2016; Wu et al., 2022). Meta-analytic findings further indicate that positive parenting practices, including authoritative parenting, are negatively associated with victim and bully-victim involvement, whereas negative parenting practices, including overprotection, are positively associated with such involvement (Lereya et al., 2013). However, another meta-analysis reported that authoritative and overprotective parenting were not significantly associated with bullying perpetration

or victimization (Nocentini et al., 2019). These mixed findings highlight the need for more longitudinal research to clarify the prospective associations between specific parenting behaviors and adolescent peer problems.

The dual-process model further emphasizes that psychological control and autonomy support may have distinct effects on positive and negative adolescent outcomes (Vansteenkiste & Ryan, 2013). Distinguishing these aspects allows researchers to examine whether one dimension is associated with outcomes independently of the other. Meta-analytic structural equation modeling analyses indicate that higher psychological control more strongly predicts higher maladaptive outcomes, whereas lower autonomy support has a stronger effect on poorer adaptive outcomes, though cross-effects also emerge (Bradshaw et al., 2025). This pattern suggests that bullying perpetration and victimization, as peer maladaptive outcomes, may be more strongly related to higher psychological control than to lower autonomy support, underscoring the need to examine both behaviors within a unified framework. However, it remains unclear whether this differential pattern extends to their mediating roles in the associations between mothers' parental burnout and adolescent bullying perpetration and victimization. Existing evidence suggests a potential asymmetry between these mediating pathways, underscoring the importance of comparing their relative strengths.

Contextual Factors

Beyond the proposed mediating processes, prior literature has identified several contextual factors linked to both parenting behaviors and adolescent bullying perpetration and victimization. First, gender shapes parenting behaviors (Zheng & Chen, 2025). Research has reported higher levels of psychological control toward boys than girls, though no significant differences have emerged for autonomy support (Chen et al., 2019, 2025). Beyond parenting behaviors, gender has also been linked to bullying perpetration and victimization, with boys generally reporting higher rates of both (Smith et al., 2019). Second, school context shapes both parenting behaviors and bullying involvement. Current emphasis on family–school collaboration reflects the view that parenting processes do not operate in isolation but interact with the school environment, jointly shaping adolescent development (Bronfenbrenner & Ceci, 1994). Schools are also the primary setting in which bullying occurs, and adolescent bullying involvement is particularly susceptible to school-level environmental influences (Hong & Espelage, 2012; Marchante et al., 2022). Third, higher family SES has been associated with less psychological control and greater autonomy support (Luo et al., 2026), as well as fewer experiences of bullying perpetration and victimization among adolescents (Tippett & Wolke, 2014). Taken together, gender, school, and SES may serve as contextual factors that shape both parenting processes and adolescent bullying involvement.

The Importance of Adolescence

Adolescence is a critical period marked by rapid biological, psychological, and social changes (Dahl et al., 2018), during which interpersonal relationships become a central developmental task in psychosocial adjustment (Erikson, 1950). Within this developmental context, adolescents increasingly explore social roles and seek status within peer circles (Hong & Espelage, 2012; Pellegrini, 2002). These developmental changes make peer dynamics particularly important for adolescents' social adjustment (Laursen & Veenstra, 2021). Moreover, adolescents increasingly seek autonomy and independence, prompting parents to adjust their roles and parenting behaviors accordingly (Sebastian et al., 2008; Wen et al., 2023). However, adapting to these changes may require greater investment of psychological, social, or material resources, thereby increasing the risk of parental burnout among parents of adolescents (Vigouroux & Scola, 2018). Given that parental burnout reflects a chronic state of resource depletion, its influence on parenting behaviors accumulates progressively over months; similarly, adolescents require time to internalize shifts in parenting before these changes extend beyond the family context into their peer interactions. Therefore, adolescence provides an important developmental window for testing these longitudinal associations. Additionally, the 12-month design (with 6-month intervals) serves as a developmentally appropriate timeframe for capturing this unfolding family-to-peer process.

Current Study

Although previous research links parental and family risk factors to adolescent bullying perpetration and victimization, whether mothers' parental burnout predicts such peer problems over time remains unclear. Furthermore, few longitudinal studies have examined higher psychological control and lower autonomy support as simultaneous mediators of this family-to-peer process. To address these gaps, the current study used three-wave longitudinal data across 12 months. It examined whether mothers' parental burnout at baseline (T1) predicted adolescents' perceptions of psychological control and autonomy support at the 6-month follow-up (T2), which then predicted bullying perpetration and victimization at the 12-month follow-up (T3). The analyses controlled for autoregressive effects and covariates,

including gender, school, and SES. This study tested the following hypotheses. Mothers' parental burnout at T1 would positively predict bullying perpetration and victimization at T3 (Hypothesis 1). Mothers' parental burnout at T1 would positively predict psychological control at T2 (Hypothesis 2a). It would negatively predict autonomy support at T2 (Hypothesis 2b). Psychological control at T2 would positively predict bullying perpetration and victimization at T3 (Hypothesis 3a). Autonomy support at T2 would negatively predict bullying perpetration and victimization at T3 (Hypothesis 3b). Psychological control at T2 would mediate the associations between mothers' parental burnout at T1 and bullying perpetration and victimization at T3 (Hypothesis 4a). Autonomy support at T2 would also mediate these associations (Hypothesis 4b). The indirect effects through psychological control at T2 would be stronger than those through autonomy support at T2 (Hypothesis 5).

Methods

Participants

This study used three waves of data (T1–T3) collected across 12 months between 2021 and 2022, with 6-month intervals between assessments. Participants were recruited from two urban schools in Hebei and Shandong Provinces, China. School 1 in Hebei Province was a private middle school in a relatively less economically developed city, whereas School 2 in Shandong Province was a public middle school in a relatively more economically developed city. A total of 1,969 adolescents and their mothers completed the baseline (T1) data collection, and adolescents participated in the T2 and T3 assessments. At T2 and T3, respectively, 65 and 241 adolescents did not complete the surveys, primarily due to school transfers or health-related absences. Ultimately, 1,697 participants completed all three waves of measurement, representing 86.20% of the baseline sample (N = 1,969). The baseline sample was retained in the subsequent analyses. In the baseline sample, 49.90% of the adolescents were boys, with a mean age of 14.35 years (SD = 1.30), and mothers had a mean age of 41.68 years (SD = 5.24). In addition, 1,834 families were two-parent families, 49 were single-parent families, 57 had other marital arrangements, and 29 had missing data on this information.

Procedure

Data were derived from a longitudinal study examining the development of adolescent mental health. The Research Ethics Committee of the Collaborative Innovation Center of Assessment for Basic Education Quality at Beijing Normal University approved the study (Ethics Approval No. 2020–2020). After the schools granted permission, homeroom teachers assisted in coordinating student participation within their classes. Prior to completing the questionnaires, both adolescents and their parents were informed about the study's purpose and provided informed consent. The research team clearly informed participants that participation was entirely voluntary and that they could withdraw at any time without penalty. Adolescents completed paper-and-pencil questionnaires in a regular classroom environment at school. The research team assigned each adolescent a unique identification code and asked adolescents to record this code and share it with their mothers after school. Homeroom teachers sent the electronic questionnaire to mothers and also provided each mother with her child's assigned identification code. Accordingly, mothers could obtain the identification code either from the homeroom teacher or from their child. This procedure ensured accurate matching between mothers and adolescents and across waves.

Measures

Parental burnout (T1)

Mothers completed the 5-item Brief Parental Burnout Scale to report their level of parental burnout (Aunola et al., 2021). An example item is "I feel like I can't take any more as a parent." Mothers rated how frequently they experienced the situation described in each statement on a 7-point Likert scale ranging from 1 (*Never*) to 7 (*Daily*), with higher scores indicating greater levels of parental burnout. The reliability of this brief version scale has been established among Chinese parents (Li et al., 2025). In the current study, the scale demonstrated good internal consistency (Cronbach's α at T1 = 0.81).

Psychological control (T1 and T2)

Adolescents completed the revised 18-item Psychological Control Scale to report their perceptions of psychological control (Wang et al., 2007). The scale comprises three subscales: guilt induction (10 items, e.g., "My parents tell me that I should feel guilty when I do not meet their expectations."), love withdrawal (5 items, e.g., "My parents act cold and unfriendly if I do something they do not like."), and authority assertion (3 items, e.g., "My parents tell me that what they want me to do is the best for me and I should not question it."). Adolescents rated each item on a 5-point scale ranging from 1 (*Strongly disagree*) to 5 (*Strongly agree*), with higher scores reflecting greater perceived parental psychological control. The scale has demonstrated good reliability among Chinese adolescents (Luo et al., 2026). In the

current study, the scale demonstrated good internal consistency across two waves (Cronbach's $\alpha = 0.92$ at T1 and 0.94 at T2).

Autonomy support (T1 and T2)

Adolescents completed the 12-item Perceived Parental Autonomy Support Scale to report their perceptions of autonomy support (Mageau et al., 2015). The scale comprises three subscales: rationale (4 items, e.g., "When I asked why I had to do, or not do, something, my parents gave me good reasons."), choice (4 items, e.g., "My parents hoped that I would make choices that corresponded to my interests and preferences regardless of what theirs were."), and acknowledgment of feelings (4 items, e.g., "My parents were able to put themselves in my shoes and understand my feelings."). Adolescents rated each item on a 5-point scale ranging from 1 (*Strongly disagree*) to 5 (*Strongly agree*), with higher scores reflecting greater perceived parental autonomy support. The scale has demonstrated good reliability among Chinese adolescents (Yang et al., 2024). In the current study, the scale demonstrated good internal consistency across two waves (Cronbach's $\alpha = 0.94$ at T1 and 0.95 at T2).

Bullying perpetration (T2 and T3)

Adolescents completed the revised 12-item Olweus Bully/Victim Questionnaire to report their experiences of bullying perpetration (Zhang et al., 1999). The bullying perpetration subscale consists of 6 items across three forms: verbal perpetration (2 items; e.g., "I called another student mean names, made fun of, or teased."), relational perpetration (2 items; e.g., "I kept him/her out of things on purpose, excluded from our group of friends, or completely ignored."), and physical perpetration (2 items; e.g., "I hit, kicked, pushed, or shoved him/her, or damaged his/her belongings in this semester."). Adolescents rated each item on a 5-point scale ranging from 1 (*Never*) to 5 (*Several times a week*), with higher scores indicating greater involvement in bullying perpetration. This questionnaire has demonstrated good reliability among Chinese adolescents (Li et al., 2026). In the current study, the bullying perpetration subscale yielded good internal consistency (Cronbach's $\alpha = 0.81$ at T2 and 0.98 at T3).

Bullying victimization (T2 and T3)

Adolescents completed the revised 12-item Olweus Bully/Victim Questionnaire to report their experiences of bullying victimization (Zhang et al., 1999). The bullying victimization subscale consists of 6 items across three forms: verbal victimization (2 items; e.g., "I was called mean names, made fun of, or teased by another student."), relational victimization (2 items; e.g., "I was kept out of things on purpose, excluded from their group of friends, or completely ignored."), and physical victimization (2 items; e.g., "I was hit, kicked, pushed, or shoved, or my belongings were damaged in this semester."). Adolescents rated each item on a 5-point scale ranging from 1 (*Never*) to 5 (*Several times a week*), with higher scores indicating greater involvement in bullying victimization. This questionnaire has demonstrated good reliability among Chinese adolescents (Li et al., 2026). In the current study, the bullying victimization subscale yielded good internal consistency (Cronbach's $\alpha = 0.81$ at T2 and 0.95 at T3).

Demographic and control variables

Adolescents reported their age, gender, school, and SES. The MacArthur Scale assessed the level of SES (Adler et al., 2000), on which adolescents selected one rung on a 10-rung ladder to indicate their family's relative standing in society, based on parents' income, education, and occupation. Scores ranged from 1 to 10, with higher scores indicating higher SES. Mothers reported their age and marital status. Based on contextual factors associated with parenting behaviors and bullying perpetration and victimization, gender, school, and SES were included as control variables.

Statistical Analyses

Preliminary analyses, including missing data analysis, common method variance test, descriptive statistics, correlations, and independent-samples *t*-tests, were conducted in SPSS. The hypothesized structural equation modeling was then estimated in Mplus. Little's missing completely at random (MCAR) test was conducted to further examine the pattern of missing data, which indicated that the missing data on the primary study variables satisfied the missing completely at random assumption ($\chi^2 = 13.661$, $df = 17$, $p = 0.691$). Moreover, no significant differences in the primary study variables were observed between participants who provided data across all three waves and those with missing data at T2 or T3 ($ps > 0.05$). These results suggest that missing data handling would introduce minimal bias; therefore, the baseline sample ($N = 1{,}969$) was retained, and missing data were addressed using the full information maximum likelihood method (FIML).

In the structural equation modeling, mothers' parental burnout was modeled as a latent construct with its five items as indicators. Psychological control, autonomy support, bullying perpetration, and bullying victimization were modeled as latent variables, using their respective subdimension scores as observed indicators. Autoregressive effects were

controlled to account for the temporal stability of variables when estimating longitudinal associations, including psychological control, autonomy support, bullying perpetration, and bullying victimization across adjacent waves. In addition, all endogenous variables (i.e., psychological control at T2, autonomy support at T2, bullying perpetration and victimization at T3) were regressed on gender, school, and SES as covariates to control for their potential confounding influence. Mediation effects were tested using a bootstrapping procedure with 5,000 resamples; indirect effects were deemed statistically significant when the 95% bias-corrected bootstrap confidence intervals excluded zero.

Several sensitivity analyses evaluated the robustness of the findings. First, alternative estimators were used to assess whether the results remained consistent across estimation methods. The main analyses used maximum likelihood estimation, which supports bootstrap confidence intervals for indirect effects. Because several study variables deviated from normality, the model was re-estimated using maximum likelihood estimation with robust standard errors. However, this estimator does not provide mediation test results or accommodate bootstrap resampling for indirect effects. Therefore, it was not used as the primary estimator in the main analyses. Bayesian estimation was then used as a complementary approach, yielding 95% credible intervals for both direct paths and mediation effects. Second, models without control variables assessed the potential impact of control-variable inclusion on the model estimates. Third, separate models for bullying perpetration and victimization assessed potential bias in model estimation attributable to restricted variance and redundancy arising from the high correlation between the two constructs.

Results

Preliminary Analyses

All adolescent-reported items were incorporated into a principal component analysis. A total of 12 components were extracted, with the first component accounting for 23.07% of the total variance, well below the 50% criterion (Podsakoff et al., 2003), suggesting limited common method bias in this study.

Table 1 presents the results of independent-samples *t*-tests for the primary variables. Regarding gender differences, boys scored significantly higher than girls on psychological control at T1, bullying perpetration at T2 and T3, and bullying victimization at T2 and T3 ($ps < 0.05$), with small to medium effect sizes (|Cohen's d| = 0.13–0.29). Regarding school differences, all variables showed statistically significant differences between the two schools ($ps < 0.05$), with small to medium effect sizes (|Cohen's d| = 0.09–0.36). Specifically, participants from School 1 scored significantly higher on mothers' parental burnout at T1, psychological control at T1, bullying perpetration at T2 and T3, and bullying victimization at T3. Participants from School 2 scored significantly higher on autonomy support at T1 and T2, psychological control at T2, and bullying victimization at T2.

Table 2 presents the means, standard deviations, skewness, kurtosis, and correlations among primary variables. Results indicated that SES was significantly negatively associated with psychological control at T1 and T2 ($r = -0.07$, $p = 0.001$; $r = -0.05$, $p = 0.031$) and significantly positively associated with autonomy support at T1 and T2 ($r = 0.18$, $p < 0.001$; $r = 0.17$, $p < 0.001$). SES was also significantly negatively associated with bullying perpetration at T3 and bullying victimization at T3 ($r = -0.08$, $p = 0.001$; $r = -0.08$, $p = 0.001$). Furthermore, mothers' parental burnout at T1 was significantly positively associated with psychological control at T2, bullying perpetration at T3, and bullying victimization at T3 ($r = 0.18$, 0.11, and 0.13, respectively; $ps < 0.001$), and significantly negatively associated with autonomy support at T2 ($r = -0.16$, $p < 0.001$). Psychological control at T2 was significantly positively associated with bullying perpetration at T3 and bullying victimization at T3 ($r = 0.10$ and 0.15, respectively; $ps < 0.001$). Autonomy support at T2 was significantly negatively associated with bullying perpetration at T3 ($r = -0.07$, $p = 0.004$) and bullying victimization at T3 ($r = -0.09$, $p < 0.001$). Overall, the correlations were consistent with the hypothesized model, providing preliminary support for the proposed prospective associations among mothers' parental burnout, psychological control, autonomy support, bullying perpetration, and bullying victimization. Additionally, several bullying perpetration and victimization variables did not meet the criteria of skewness below 3 and kurtosis below 10 (Kline, 2016), whereas the other variables met these standards. Sensitivity analyses were conducted using maximum likelihood estimation with robust standard errors to evaluate the robustness of the results.

Structural Equation Modeling Analyses

The hypothesized structural model demonstrated acceptable fit to the data: comparative fit index (CFI) = 0.934, Tucker-Lewis index (TLI) = 0.924, root mean square error of approximation (RMSEA) = 0.058. All fit indices satisfied the recommended criteria (CFI and TLI ≥ 0.900, RMSEA ≤ 0.080; Wen et al., 2004), indicating that the model adequately represented the data. Additionally, Table 3 reports standardized loadings for all observed variables.

Autoregressive effects indicated that psychological control at T1 predicted psychological control at T2 ($\beta = 0.481$,

Table 1 Differences in primary variables by gender and school

Variables	Groups	*M*	*SD*	*t*	*p*	*d*
T1 Mothers' parental burnout	Boys	1.23	0.35	1.18	0.237	0.05
	Girls	1.21	0.35			
T1 Psychological control	Boys	2.62	0.81	4.58***	<0.001	0.21
	Girls	2.45	0.80			
T1 Autonomy support	Boys	3.86	0.84	1.72	0.086	0.08
	Girls	3.80	0.77			
T2 Psychological control	Boys	2.19	0.88	-0.02	0.987	0.00
	Girls	2.19	0.87			
T2 Autonomy support	Boys	3.71	0.93	1.69	0.092	0.08
	Girls	3.64	0.86			
T2 Bullying perpetration	Boys	1.14	0.31	3.83***	<0.001	0.18
	Girls	1.09	0.29			
T2 Bullying victimization	Boys	1.30	0.55	2.92**	0.004	0.13
	Girls	1.23	0.47			
T3 Bullying perpetration	Boys	1.29	0.71	5.98***	<0.001	0.29
	Girls	1.12	0.43			
T3 Bullying victimization	Boys	1.37	0.76	3.66***	<0.001	0.18
	Girls	1.25	0.59			
T1 Mothers' parental burnout	School 1	1.24	0.36	3.17**	0.002	0.14
	School 2	1.19	0.33			
T1 Psychological control	School 1	2.59	0.75	3.05**	0.002	0.14
	School 2	2.48	0.87			
T1 Autonomy support	School 1	3.69	0.72	-7.95***	<0.001	-0.36
	School 2	3.98	0.87			
T2 Psychological control	School 1	2.14	0.80	-2.30*	0.021	-0.11
	School 2	2.24	0.95			
T2 Autonomy support	School 1	3.63	0.80	-2.40*	0.016	-0.11
	School 2	3.73	0.98			
T2 Bullying perpetration	School 1	1.13	0.32	2.39*	0.017	0.11
	School 2	1.10	0.28			
T2 Bullying victimization	School 1	1.24	0.47	-2.02*	0.044	-0.09
	School 2	1.29	0.55			
T3 Bullying perpetration	School 1	1.26	0.66	4.55***	<0.001	0.22
	School 2	1.13	0.51			
T3 Bullying victimization	School 1	1.36	0.73	3.32***	<0.001	0.16
	School 2	1.25	0.61			

Note. School 1 in Hebei Province was located in a relatively less economically developed city; School 2 in Shandong Province was located in a relatively more economically developed city

*$p<0.05$, **$p<0.01$, ***$p<0.001$

95% CI [0.420, 0.541]), and autonomy support at T1 predicted autonomy support at T2 ($\beta=0.441$, 95% CI [0.378, 0.502]). Similarly, bullying perpetration at T2 predicted bullying perpetration at T3 ($\beta=0.094$, 95% CI [0.055, 0.140]), and bullying victimization at T2 predicted bullying victimization at T3 ($\beta=0.238$, 95% CI [0.165, 0.319]). Regarding control variables, gender (1 = boys and 2 = girls) was positively associated with psychological control at T2 ($\beta=0.049$, 95% CI [0.007, 0.091]). Gender was negatively associated with both bullying perpetration at T3 ($\beta=-0.143$, 95% CI [−0.184, −0.096]) and bullying victimization at T3 ($\beta=-0.088$, 95% CI [−0.134, −0.039]), whereas its association with autonomy support at T2 was not significant ($\beta=-0.042$, 95% CI [−0.083, 0.001]). School (1 = school in Hebei Province, China, and 2 = school in Shandong Province, China) was positively associated with psychological control at T2 ($\beta=0.125$, 95% CI [0.081, 0.170]) and negatively associated with autonomy support at T2 ($\beta=-0.083$, 95% CI [−0.130, −0.037]), bullying perpetration at T3 ($\beta=-0.097$, 95% CI [−0.150, −0.043]), and bullying victimization at T3 ($\beta=-0.082$, 95% CI [−0.135, −0.029]). In addition, SES negatively predicted psychological control at T2 ($\beta=-0.058$, 95% CI [−0.110, −0.006]) and positively predicted autonomy support at T2 ($\beta=0.127$, 95% CI [0.076, 0.178]).

Table 2 Descriptive statistics and correlations among primary variables

Variables	1	2	3	4	5	6	7	8	9	10	11	12
1. Adolescents' age	—											
2. Mothers' age	0.19***	—										
3. SES	-0.18***	0.00	—									
4. T1 Mothers' parental burnout	0.01	-0.01	-0.04	—								
5. T1 Psychological control	0.02	-0.02	-0.07**	0.21***	—							
6. T1 Autonomy support	-0.04	0.00	0.18***	-0.21***	-0.34***	—						
7. T2 Psychological control	-0.04	-0.02	-0.05*	0.18***	0.46***	-0.36***	—					
8. T2 Autonomy support	-0.03	-0.01	0.17***	-0.16***	-0.30***	0.47***	-0.45***	—				
9. T2 Bullying perpetration	0.05*	-0.02	-0.04	0.03	0.07**	-0.11***	0.16***	-0.18***	—			
10. T2 Bullying victimization	-0.05*	0.01	-0.02	0.11***	0.18***	-0.24***	0.29***	-0.23***	0.35***	—		
11. T3 Bullying perpetration	0.10***	0.02	-0.08**	0.11***	0.12***	-0.11***	0.10***	-0.07**	0.16***	0.10***	—	
12. T3 Bullying victimization	0.07**	0.02	-0.08**	0.13***	0.13***	-0.12***	0.15***	-0.09***	0.16***	0.27***	0.85***	—
Skewness	1.26	0.59	0.13	1.91	0.41	-0.75	0.73	-0.66	5.85	3.37	3.56	2.81
Kurtosis	0.13	0.11	1.06	3.72	-0.15	0.60	0.09	0.28	49.74	14.61	13.77	8.47
M	14.35	41.68	5.00	1.22	2.54	3.83	2.19	3.68	1.11	1.27	1.20	1.31
SD	1.30	5.24	1.51	0.35	0.81	0.81	0.87	0.89	0.30	0.51	0.59	0.68

Note. $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$

Table 3 Standardized loadings for all latent variables

Variables	Items/Factors	Standardized loadings
T1 Mothers' parental burnout	Item 1	0.705***
	Item 2	0.748***
	Item 3	0.623***
	Item 4	0.627***
	Item 5	0.672***
T1 Psychological control	Guilt induction	0.901***
	Love withdrawal	0.783***
	Authority assertion	0.737***
T1 Autonomy support	Rationale	0.847***
	Choice	0.884***
	Acknowledgment of feelings	0.933***
T2 Psychological control	Guilt induction	0.948***
	Love withdrawal	0.787***
	Authority assertion	0.726***
T2 Autonomy support	Rationale	0.894***
	Choice	0.896***
	Acknowledgment of feelings	0.951***
T2 Bullying perpetration	Verbal perpetration	0.689***
	Relational perpetration	0.837***
	Physical perpetration	0.822***
T2 Bullying victimization	Verbal victimization	0.785***
	Relational victimization	0.820***
	Physical victimization	0.704***
T3 Bullying perpetration	Verbal perpetration	0.974***
	Relational perpetration	0.979***
	Physical perpetration	0.971***
T3 Bullying victimization	Verbal victimization	0.921***
	Relational victimization	0.902***
	Physical victimization	0.938***

Note. $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$

After controlling for autoregressive and demographic variables, the results indicated that mothers' parental burnout at T1 significantly positively predicted both bullying perpetration at T3 ($\beta=0.096$, 95% CI [0.031, 0.163]) and bullying victimization at T3 ($\beta=0.099$, 95% CI [0.039, 0.166]), suggesting that higher mothers' parental burnout was associated with greater subsequent bullying perpetration and victimization. Mothers' parental burnout at T1 also significantly positively predicted psychological control at T2 ($\beta=0.102$, 95% CI [0.043, 0.163]) and significantly negatively predicted autonomy support at T2 ($\beta = -0.095$, 95% CI [−0.150, −0.040]), suggesting that higher mothers' parental burnout was associated with greater psychological control and lower autonomy support over time. Furthermore, psychological control at T2 significantly positively predicted both bullying perpetration at T3 ($\beta=0.080$, 95% CI [0.018, 0.143]) and bullying victimization at T3 ($\beta=0.079$, 95% CI [0.016, 0.140]), suggesting that higher psychological control was associated with increased subsequent bullying perpetration and victimization. In contrast, autonomy

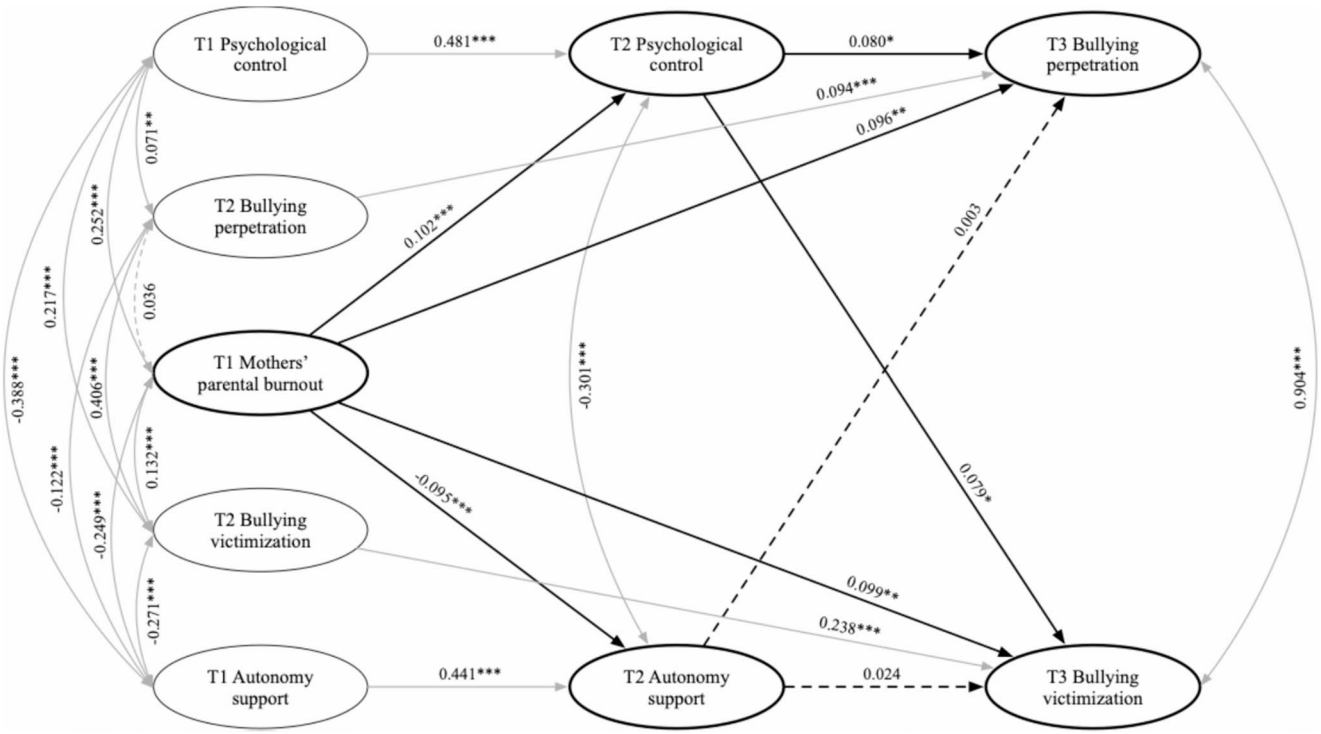


Fig. 1 Standardized path coefficients. Note For clarity, control variables (gender, school, SES) are not depicted in the figure, where solid lines represent significant paths, and dashed lines indicate non-significant paths. $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$

Table 4 Mediation effect tests

Pathway	Standardized effect
T3 Bullying perpetration	
T1 Mothers' parental burnout→T3 Bullying perpetration (Total)	**0.104 [0.040, 0.168]**
T1 Mothers' parental burnout→T3 Bullying perpetration (Direct)	**0.096 [0.031, 0.163]**
T1 Mothers' parental burnout→T2 Psychological control→T3 Bullying perpetration	**0.008 [0.002, 0.019]**
T1 Mothers' parental burnout→T2 Autonomy support→T3 Bullying perpetration	0.000 [-0.007, 0.005]
T3 Bullying victimization	
T1 Mothers' parental burnout→T3 Bullying victimization (Total)	**0.104 [0.045, 0.170]**
T1 Mothers' parental burnout→T3 Bullying victimization (Direct)	**0.099 [0.039, 0.166]**
T1 Mothers' parental burnout→T2 Psychological control→T3 Bullying victimization	**0.008 [0.002, 0.018]**
T1 Mothers' parental burnout→T2 Autonomy support→T3 Bullying victimization	-0.002 [-0.010, 0.003]

Note. Bold values indicate statistically significant effects (95% CI does not include zero)

support at T2 did not significantly predict either bullying perpetration at T3 (β=0.003, 95% CI [−0.056, 0.064]) or bullying victimization at T3 (β=0.024, 95% CI [−0.038, 0.088]), suggesting that autonomy support was not associated with subsequent bullying perpetration and victimization. Figure 1 shows the standardized path coefficients.

Further mediation analyses examined the roles of psychological control at T2 and autonomy support at T2 in the associations between mothers' parental burnout at T1 and both bullying perpetration at T3 and bullying victimization at T3. The results showed that psychological control, rather than autonomy support, mediated the longitudinal associations between mothers' parental burnout at T1 and both bullying perpetration and victimization at T3 (see details in Table 4). To examine the relative importance of psychological control at T2 and autonomy support at T2 as mediators, a test of indirect effects was conducted. The results indicated that in the pathway from mothers' parental burnout at T1 to bullying perpetration at T3, the indirect effect through psychological control at T2 was greater than that through autonomy support at T2, and this difference was significant at the 90% CI. In the pathway from mothers' parental burnout at T1 to bullying victimization at T3, the indirect effect through psychological control at T2 was greater than that through autonomy support at T2, and this difference was significant at the 95% CI.

Sensitivity Analyses

First, to assess the robustness of the findings to the choice of estimator, sensitivity analyses re-estimated the model using maximum likelihood estimation with robust standard errors and Bayesian estimation. The direction and

inferential conclusions of all path estimates and indirect effects remained consistent across estimators (see Table S1 and Table S2 in the supplemental materials).

Second, to assess the potential impact of control variable inclusion on model estimation, the model was re-estimated after removing all control variables. The pattern of results remained consistent with that of the main analyses, indicating that the inclusion of control variables did not affect the findings (see Table S1 and Table S2 in the supplemental materials).

Third, to assess potential bias in model estimation attributable to restricted variance and item redundancy arising from the high intercorrelation between bullying perpetration and victimization, the sensitivity analyses estimated separate models. In these models, bullying perpetration and victimization each served independently as the outcome variable. The results remained consistent with those of the main analyses, suggesting that the high intercorrelation did not introduce meaningful bias in path estimation (see Table S1 and Table S2 in the supplemental materials). Notably, the main analyses modeled bullying perpetration and victimization simultaneously as joint outcomes to retain the covariance structure and minimize the risk of inflated Type II error.

Discussion

Previous research has provided limited insight into whether mothers' parental burnout contributes to downstream peer outcomes and the underlying longitudinal mechanisms. Using three waves of data spanning 12 months, this study aimed to examine the longitudinal associations between mothers' parental burnout at baseline (T1) and adolescent bullying perpetration and victimization at the 12-month follow-up (T3). It also tested the differential mediating roles of adolescents' perceptions of psychological control and autonomy support at the 6-month follow-up (T2) within the same framework. After controlling for gender, school, SES, and autoregressive effects, mothers' parental burnout at T1 predicted higher levels of adolescent bullying perpetration and victimization at T3. These associations were mediated through higher psychological control at T2, rather than through lower autonomy support at T2. This asymmetric mediation pattern integrates and extends both the parenting process model and the dual-process model and has longitudinal implications for understanding the parenting mechanisms underlying adolescent bullying perpetration and victimization.

Preliminary analyses supported including gender, school, and SES as covariates in the longitudinal models. Consistent with prior research (Shek, 2008), boys reported higher psychological control at T1 than girls, a pattern that may reflect parents' higher expectations for boys. While prior research suggested that this gender difference may be relatively stable across multiple waves (Shek, 2008), the current study showed that it did not persist at T2, possibly indicating adolescents' increasing autonomy needs over time. Boys also reported higher bullying perpetration and victimization at T2 and T3, a pattern consistent with prior research (Smith et al., 2019) and potentially attributable to the heightened competition, dominance, and overt aggression characteristic of boys' peer dynamics. In addition, participants from School 1, located in a relatively less economically developed city, reported higher levels of mothers' parental burnout and psychological control at T1. They also reported higher bullying perpetration at T2 and T3 and higher bullying victimization at T3. This pattern suggests that regional economic disadvantage may place additional strain on parents and adolescents. Finally, adolescents with lower SES reported higher psychological control at T1 and T2, lower autonomy support at T1 and T2, and higher bullying perpetration and victimization at T3. These associations suggest that limited family resources may constitute a developmental risk context.

After accounting for contextual factors (gender, school, and SES) and autoregressive effects, the current study provides longitudinal evidence that mothers' parental burnout at T1 predicts higher levels of both adolescent bullying perpetration and victimization at T3. These findings align with prior research linking parental burnout to maladaptive developmental outcomes, including antisocial behavior and other externalizing problems (Chen et al., 2022; Cheng et al., 2020). More importantly, the present findings advance the literature by showing that the adverse implications of parental burnout are not confined to adolescents' individual maladjustment (e.g., internalizing and externalizing problems; see Chen et al., 2022) but may also extend to maladaptive functioning in peer contexts, specifically bullying perpetration and victimization. This distinction is theoretically meaningful, as bullying perpetration and victimization are inherently interpersonal phenomena shaped by relational dynamics beyond the individual. Overall, these findings support and extend the social-ecological model of bullying (Swearer & Hymel, 2015) by identifying parental burnout as an important resource-deficient parental and family risk factor that may contribute not only to bullying perpetration but also to bullying victimization.

Mothers' parental burnout at T1 was associated with higher adolescent bullying perpetration and victimization at T3 through higher psychological control at T2, consistent with the assumptions of the parenting process model (Belsky, 1984; Taraban & Shaw, 2018) and prior empirical findings (Peng et al., 2024; Ren et al., 2023; Zhang et

al., 2025). Parental resources are an important determinant of parenting quality: as parental burnout intensifies, emotionally exhausted parents may adopt psychological control strategies as a low-cost means of managing adolescent behavior, due to limited cognitive and emotional resources. Current longitudinal evidence further indicates that this burnout-driven increase in psychological control contributes to adolescents' maladaptive functioning in peer contexts, specifically bullying perpetration and victimization. Specifically, parental psychological control suppresses adolescents' emotional expression, thereby depriving them of opportunities to develop emotion regulation skills within the family context (Beliveau et al., 2023). As a result, when navigating interpersonal difficulties with peers, adolescents with limited emotion regulation capacities are more likely to rely on maladaptive strategies, which contribute to both bullying perpetration and victimization (Bäker et al., 2023).

Although mothers' parental burnout at T1 predicted lower autonomy support at T2, autonomy support at T2 did not significantly predict adolescent bullying perpetration and victimization at T3. Therefore, the indirect effects through autonomy support at T2 were not significant. Notably, mothers' parental burnout was associated with lower autonomy support, suggesting that mothers experiencing parental burnout undergo a qualitative shift in parenting orientation, moving away from autonomy-supportive practices and toward more controlling practices. However, the link between autonomy support and bullying perpetration and victimization in the current study diverges from some previous findings, which generally reported that lower autonomy support is associated with higher levels of bullying perpetration and victimization (Rajendran et al., 2016; Wu et al., 2022). This discrepancy may be better understood within the dual-process model (Vansteenkiste & Ryan, 2013). The current study simultaneously examined both psychological control and autonomy support within the same model, controlling for their mutual covariance, as well as the autoregressive effects and additional covariates. This approach may help capture parenting dynamics more realistically. Existing research on simultaneous models of psychological control, autonomy support, adaptive outcomes, and maladaptive outcomes has shown that psychological control exerts a stronger influence on maladaptive outcomes, whereas autonomy support more strongly predicts adaptive outcomes (Bradshaw et al., 2025). Additional analyses further showed that the indirect pathway through psychological control at T2 was stronger than that through autonomy support at T2 for both outcomes. This difference reached significance at the 90% CI for bullying perpetration and at the 95% CI for bullying victimization. Related evidence also found that psychological control and autonomy support can both mediate links between parental burnout and adolescents' emotional and behavioral problems, but the indirect effect via autonomy support may be considerably weaker than that via psychological control (Yan et al., 2026).

These findings underscore the importance of considering both psychological control and autonomy support concurrently. Moreover, as emphasized by the dual-process model (Vansteenkiste & Ryan, 2013), the protective effects of positive parenting on negative outcomes are often less direct than the detrimental effects of negative parenting and may operate through additional mediating mechanisms. For instance, reductions in autonomy support may particularly contribute to adolescent self-doubt, insecure attachment, and difficulties in trusting others (Landry et al., 2008; Lerner & Steinberg, 2009; Van Petegem et al., 2018). Individuals' internal schemas, such as representations of safety, trust, power, esteem, and intimacy, are further linked to interpersonal outcomes, including relationship difficulties, revictimization, and the victim-to-victimizer cycle (McCann et al., 1988).

Strengths and Implications

The three-wave longitudinal design used 6-month intervals and controlled for autoregressive effects and relevant covariates, thereby strengthening the basis for longitudinal inference regarding the downstream associations of mothers' parental burnout. More specifically, the study linked mothers' parental burnout at baseline to adolescent bullying perpetration and victimization at the 12-month follow-up, while also advancing understanding of parental and family risks underlying these outcomes. Furthermore, this study simultaneously examined adolescents' perceptions of psychological control and autonomy support as potential mediators at the 6-month follow-up and revealed an asymmetric pattern. In addition, relevant findings demonstrated robustness across a series of sensitivity analyses. Together, these findings clarify how intra-familial dynamics cascade into extra-familial peer outcomes within 12 months, while also integrating and extending existing theories and informing practical implications.

Theoretically, the findings extend the social-ecological model of bullying by identifying mothers' parental burnout as a parental and family risk factor in adolescents' later bullying perpetration and victimization. Moreover, the findings integrate and extend the parenting process model and the dual-process model by showing an asymmetric mediation pattern. When psychological control and autonomy support were examined simultaneously, psychological control, rather than autonomy support, mediated the prospective associations between mothers' parental burnout and adolescent bullying perpetration and victimization over time. Distinguishing between negative and positive parenting behaviors within a longitudinal framework is not only a

methodological refinement but also strengthens theoretical understanding. The processes by which negative parenting behaviors transmit risk to peer outcomes differ from those through which positive parenting behaviors offer protection. Psychological control may directly undermine adolescents' emotion regulation and social functioning through coercive and intrusive mechanisms, whereas low autonomy support likely affects negative outcomes more indirectly, possibly through additional mechanisms.

Practically, the findings highlight the importance of attending to parents' psychological, social, and material resources, as well as to the maladaptive parenting behaviors that may follow from their depletion. These processes may have longitudinal implications not only for parenting practices but also for adolescents' broader social development. Furthermore, the differential roles of psychological control and autonomy support carry practical significance, suggesting that early efforts aimed at reducing psychological control may be particularly effective in disrupting the longitudinal pathway from mothers' parental burnout to adolescent bullying perpetration and victimization. In other words, reducing harmful parenting behaviors may be a more immediate and reliable target than increasing positive parenting practices (Karwatowska et al., 2025).

Limitations and Future Directions

Despite the above strengths and implications, several limitations remain and warrant consideration in future research. First, the absence of multi-wave data for some study variables limits the ability to examine reciprocal relationships. Indeed, adolescent bullying perpetration and victimization may, in turn, influence parenting behaviors and exacerbate parental burnout (Peng et al., 2024; Ren et al., 2024; Woine et al., 2026). In addition, parenting behaviors may also trigger parents' burnout experiences (Nguyen et al., 2023). These important reciprocal dynamics could not be addressed in the current study and highlight the need for further investigation in future research.

Second, the large sample size provided sufficient statistical power but may have also increased the likelihood that relatively small associations reached statistical significance. Therefore, the magnitude of the observed effects should be interpreted with caution. However, it is important to note that small path coefficients are expected in autoregressive longitudinal models because prospective effects are estimated after controlling for the prior stability of each endogenous variable. Moreover, the 12-month design captures short-term longitudinal associations and cannot determine whether the same patterns persist across longer developmental periods. Future studies should examine the robustness of these findings in independent samples and over longer intervals, such as annual assessments.

Third, the current study did not find significant associations between autonomy support and adolescent bullying perpetration or victimization, suggesting the possible presence of unexamined mediating mechanisms. Although exploring these mechanisms was beyond the scope of the current research, they warrant further investigation in future studies. Moreover, according to the dual-process model (Vansteenkiste & Ryan, 2013), autonomy support may be particularly relevant to defending behaviors in peer contexts (Iotti et al., 2023). The current study focused on adolescents' negative outcomes (i.e., bullying perpetration and victimization) within peer dynamics without examining positive outcomes (i.e., defending behaviors). Simultaneously investigating the differential effects of psychological control and autonomy support on both negative and positive peer outcomes could clarify the relations among parenting dimensions and adolescent peer dynamics. This direction is especially deserving of future research.

Fourth, this study focused exclusively on mothers' parental burnout, leaving fathers' parental burnout unexamined. Although existing evidence suggests that mothers' burnout may exert a more pronounced influence on adolescent development (Luo et al., 2025; Wang et al., 2023; Yang et al., 2023), the absence of fathers' data constrains both the comprehensiveness of the findings and broader conclusions about family-level processes. Future research should include both parents to facilitate more detailed comparisons of parental roles.

Finally, although data were collected from both mothers and adolescents, several key constructs (i.e., autonomy support, psychological control, bullying perpetration, and bullyingvictimization) were reported solely by adolescents. The single-factor test suggested that common method variance was not severe, future research could strengthen the robustness of findings by employing multi-informant or multi-method approaches. For example, parenting behaviors could be assessed via reports from both parents and adolescents (Zheng & Chen, 2025). Additionally, self-reported bullying measures may introduce specific biases, including underreporting of bullying perpetration or differential perceptions of victimization. The relatively low levels of bullying perpetration and victimization observed in this study may partly reflect self-report bias. Another possibility is that these low levels reflect a floor effect associated with the Chinese cultural emphasis on interpersonal harmony, as prior studies have also reported similar floor effects (Li et al., 2026). Future research may therefore benefit from incorporating peer-reported assessments, such as the peer nomination method. This method is widely used to capture

classroom peer dynamics (Bouman et al., 2012; Branson & Cornell, 2009).

Conclusion

Longitudinal research on the role of mothers' parental burnout in adolescent bullying perpetration and victimization remains limited, and little is known about the parenting behavior mechanisms underlying these associations. Drawing on a three-wave longitudinal design, the present study examined the longitudinal associations among mothers' parental burnout, adolescents' perceptions of psychological control and autonomy support, and adolescent bullying perpetration and victimization. The results showed that mothers' parental burnout predicted higher adolescent bullying perpetration and victimization over time, and that these associations were further mediated by increased psychological control, rather than decreased autonomy support. The indirect effect differences reached significance at the 90% CI for bullying perpetration and at the 95% CI for bullying victimization. These findings clarify how mothers' parental burnout may contribute to adolescents' later peer problems through specific parenting behaviors, particularly psychological control.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10964-026-02371-4.

Acknowledgments We sincerely thank the editors and the reviewers for their constructive comments, which have greatly improved the quality and clarity of this manuscript. We also thank the adolescents and mothers who participated in this study for their time and contributions.

Authors' Contributions HB conceived the study, participated in its design, performed statistical analysis, interpreted the data, and drafted the manuscript; QP conceived the study, participated in its design, collected the data, interpreted the data, and reviewed the manuscript; YH conceived the study, participated in its design, and reviewed the manuscript; YZ conceived the study and reviewed the manuscript; FC conceived the study, participated in its design, collected the data, coordinated and reviewed the manuscript. All authors read and approved the final manuscript.

Funding This work was supported by the Beijing Office for Education Sciences Planning [Grant Number: BAHA24026].

Data Availability The datasets generated and/or analyzed during the current study are not publicly available but are available from the corresponding author on reasonable request. The replication file for the analyses is available in OSF at https://osf.io/d5hrn.

Declarations

Ethical Approval This study was performed in line with the principles of the Declaration of Helsinki. Ethical approval was granted by the Research Ethics Committee of the Collaborative Innovation Center of Assessment for Basic Education Quality at Beijing Normal University (No. 2020-2020).

Informed Consent Informed consent was obtained from all individual participants included in the study.

Conflicts of Interest The authors declare no competing interests.

References

Adler, N. E., Epel, E. S., Castellazzo, G., & Ickovics, J. R. (2000). Relationship of subjective and objective social status with psychological and physiological functioning: Preliminary data in healthy, White women. *Health Psychology*, *19*(6), 586–592. https://doi.org/10.1037/0278-6133.19.6.586

Ariani, T. A., Putri, A. R., Firdausi, F. A., & Aini, N. (2025). Global prevalence and psychological impact of bullying among children and adolescents: A meta-analysis. *Journal of Affective Disorders*, *385*, 119446. https://doi.org/10.1016/j.jad.2025.119446

Arseneault, L. (2018). Annual research review: The persistent and pervasive impact of being bullied in childhood and adolescence: Implications for policy and practice. *Journal of Child Psychology and Psychiatry and Allied Disciplines*, *59*(4), 405–421. https://doi.org/10.1111/jcpp.12841

Aunola, K., Sorkkila, M., Tolvanen, A., Tassoul, A., Mikolajczak, M., & Roskam, I. (2021). Development and validation of the Brief Parental Burnout Scale (BPBS). *Psychological Assessment*, *33*(11), 1125–1137. https://doi.org/10.1037/pas0001064

Bäker, N., Wilke, J., Eilts, J., & von Düring, U. (2023). Understanding the complexities of adolescent bullying: The interplay between peer relationships, emotion regulation, and victimization. *New Directions for Child and Adolescent Development*, *2023,*9916294 . https://doi.org/10.1155/2023/9916294

Bandura, A. (1977). *Social learning theory*. Prentice-Hall.

Beliveau, L. E., Iselin, A. M. R., DeCoster, J., & Boyer, M. A. (2023). A meta-analysis relating parental psychological control with emotion regulation in youth. *Journal of Child and Family Studies*, *32*(12), 3876–3891. https://doi.org/10.1007/s10826-023-02700-2

Belsky, J. (1984). The determinants of parenting: A process model. *Child Development*, *55*(1), 83–96. https://doi.org/10.2307/1129836

Belsky, J., & Jaffee, S. R. (2006). The multiple determinants of parenting. In D. Cicchetti, & D. J. Cohen (Eds.), *Developmental psychopathology: Vol. 3. Risk, disorder, and adaptation* (2nd ed., pp. 38–85). Wiley.

Berkowitz, L. (1989). Frustration-aggression hypothesis: Examination and reformulation. *Psychological Bulletin*, *106*(1), 59–73. https://doi.org/10.1037/0033-2909.106.1.59

Bian, H., Bian, Y., & Liang, L. (2025). The impact of parental burnout on family cohesion: The mediating roles of parent-child relationships and parental differences. *Current Psychology*, *44*(10), 9435–9445. https://doi.org/10.1007/s12144-025-07729-8

Bian, H., Liang, L., Bian, Y., & Zhao, M. (2026a). Reciprocal relationships between bullying victimization and prosocial behavior and the mediating role of self-esteem among Chinese boys and girls: An RI-CLPM study. *Journal of Youth and Adolescence*, *55*(5), 1325–1342. https://doi.org/10.1007/s10964-026-02329-6

Bian, H., Zhuang, R., Wang, J., Bian, Y., & Liang, L. (2026b). Parental burnout and adolescents' internet gaming addiction: Roles of basic psychological need satisfaction, effortful control and gender. *Psychological Development and Education*, *42*(1), 110–120. https://doi.org/10.16187/j.cnki.issn1001-4918.2026.01.11

Bouman, T., van der Meulen, M., Goossens, F. A., Olthof, T., Vermande, M. M., & Aleva, E. A. (2012). Peer and self-reports of victimization and bullying: Their differential association with internalizing problems and social adjustment. *Journal of School Psychology*, *50*(6), 759–774. https://doi.org/10.1016/j.jsp.2012.08.004

Bradshaw, E. L., Duineveld, J. J., Conigrave, J. H., Steward, B. A., Ferber, K. A., Joussemet, M., Parker, P. D., & Ryan, R. M. (2025). Disentangling autonomy-supportive and psychologically controlling parenting: A meta-analysis of self-determination theory's dual process model across cultures. *American Psychologist*, *80*(6), 879–895. https://doi.org/10.1037/amp0001389

Branson, C. E., & Cornell, D. G. (2009). A comparison of self and peer reports in the assessment of middle school bullying. *Journal of Applied School Psychology*, *25*(1), 5–27. https://doi.org/10.1080/15377900802484133

Bronfenbrenner, U., & Ceci, S. J. (1994). Nature-nurture reconceptualized in developmental perspective: A bioecological model. *Psychological Review*, *101*(4), 568–586. https://doi.org/10.1037/0033-295X.101.4.568

Chen, Y., Li, R., & Liu, X. (2019). The relationships among parental psychological control/autonomy support, self-trouble, and internalizing problems across adolescent genders. *Scandinavian Journal of Psychology*, *60*(6), 539–547. https://doi.org/10.1111/sjop.12573

Chen, B. B., Qu, Y., Yang, B., & Chen, X. (2022). Chinese mothers' parental burnout and adolescents' internalizing and externalizing problems: The mediating role of maternal hostility. *Developmental Psychology*, *58*(4), 768–777. https://doi.org/10.1037/dev0001311

Chen, S., Jiang, Y., Qiu, S., Hu, J., Wang, L., Jiang, Y., & Wang, R. (2025). Distinct roles of parental autonomy support and psychological control in Chinese elementary school students' intelligence mindsets and academic outcomes. *The Asia-Pacific Education Researcher*, *34*(1), 133–146. https://doi.org/10.1007/s40299-024-00842-5

Cheng, H., Wang, W., Wang, S., Li, Y., Liu, X., & Li, Y. (2020). Validation of a Chinese version of the Parental Burnout Assessment. *Frontiers in Psychology*, *11*, 321. https://doi.org/10.3389/fpsyg.2020.00321

Choi, B., & Park, S. (2021). Bullying perpetration, victimization, and low self-esteem: Examining their relationship over time. *Journal of Youth and Adolescence*, *50*(4), 739–752. https://doi.org/10.1007/s10964-020-01379-8

Dahl, R. E., Allen, N. B., Wilbrecht, L., & Suleiman, A. B. (2018). Importance of investing in adolescence from a developmental science perspective. *Nature*, *554*(7693), 441–450. https://doi.org/10.1038/nature25770

Dollard, J., Miller, N. E., Doob, L. W., Mowrer, O. H., & Sears, R. R. (1939). *Frustration and aggression*. Yale University Press.

Erikson, E. H. (1950). *Childhood and society*. W. W. Norton & Company.

Fredrick, S. S., Nickerson, A. B., & Livingston, J. A. (2022). Family cohesion and the relations among peer victimization and depression: A random intercepts cross-lagged model. *Development and Psychopathology*, *34*(4), 1429–1446. https://doi.org/10.1017/S0954579421000016X

Hong, J. S., & Espelage, D. L. (2012). A review of research on bullying and peer victimization in school: An ecological system analysis. *Aggression and Violent Behavior*, *17*(4), 311–322. https://doi.org/10.1016/j.avb.2012.03.003

Iotti, N. O., Menin, D., Longobardi, C., & Jungert, T. (2023). Investigating the effects of autonomy-supportive parenting practices on Italian young adolescent students' motivation to defend victims of bullying: Findings on the mediating roles of reactance, depression, anxiety, and stress. *Frontiers in Psychology*, *14*, 1156807. https://doi.org/10.3389/fpsyg.2023.1156807

Karwatowska, L., Solmi, F., Baldwin, J. R., Jaffee, S. R., Viding, E., Pingault, J. B., & De Stavola, B. L. (2025). Positive and negative parenting practices and offspring disruptive behavior: A meta-analytic review of quasi-experimental evidence. *Psychological Bulletin*, *151*(11), 1363–1381. https://doi.org/10.1037/bul0000495

Kline, R. B. (2016). *Principles and practice of structural equation modeling* (4th ed.). The Guilford Press.

Landry, R., Whipple, N., Mageau, G., Joussemet, M., Koestner, R., DiDio, L., Gingras, I., Bernier, A., & Haga, S. M. (2008). Trust in organismic development, autonomy support, and adaptation among mothers and their children. *Motivation and Emotion*, *32*(3), 173–188. https://doi.org/10.1007/s11031-008-9092-2

Laursen, B., & Veenstra, R. (2021). Toward understanding the functions of peer influence: A summary and synthesis of recent empirical research. *Journal of Research on Adolescence*, *31*(4), 889–907. https://doi.org/10.1111/jora.12606

Lereya, S. T., Samara, M., & Wolke, D. (2013). Parenting behavior and the risk of becoming a victim and a bully/victim: A meta-analysis study. *Child Abuse & Neglect*, *37*(12), 1091–1108. https://doi.org/10.1016/j.chiabu.2013.03.001

Lerner, R. M., & Steinberg, L. (Eds.). (2009). *Handbook of adolescent psychology: Vol. 1. Individual bases of adolescent development* (3rd ed.). Wiley.

Li, Y., Wang, S., & Bian, Y. (2025). Associations between parenting stressors and parental burnout: The roles of parenting stress responses and child-based worth. *Current Psychology*, *44*(24), 19228–19241. https://doi.org/10.1007/s12144-025-08486-4

Li, J., Lu, Y., & Chu, X. (2026). Joint developmental trajectories of bullying victimization/perpetration and bystanders' defending behavior in adolescence: A latent growth mixture model. *Journal of Youth and Adolescence*, *55*(5), 1181–1201. https://doi.org/10.1007/s10964-026-02321-0

Luo, X., Wang, H., Xu, J., Liu, H., Suveg, C., & Han, Z. R. (2025). Dynamic processes of parent–adolescent conflict and warmth in Chinese families: Differences between mothers and fathers. *Journal of Youth and Adolescence*, *54*(8), 1985–1998. https://doi.org/10.1007/s10964-025-02160-5

Luo, Y., Li, R., Yang, Z., Luo, R., Yu, H., Chen, F., & Wang, Y. (2026). Joint trajectories of parental psychological control and autonomy support and their impact on Chinese adolescents' psychological adjustment. *Journal of Youth and Adolescence*, *55*(3), 753–767. https://doi.org/10.1007/s10964-025-02272-y

Mageau, G. A., Ranger, F., Joussemet, M., Koestner, R., Moreau, E., & Forest, J. (2015). Validation of the Perceived Parental Autonomy Support Scale (P-PASS). *Canadian Journal of Behavioural Science / Revue Canadienne Des Sciences Du Comportement*, *47*(3), 251–262. https://doi.org/10.1037/a0039325

Marchante, M., Coelho, A., V., & Maria Romão, A. (2022). The influence of school climate in bullying and victimization behaviors during middle school transition. *Contemporary Educational Psychology*, *71*, 102111. https://doi.org/10.1016/j.cedpsych.2022.102111

McCann, I. L., Sakheim, D. K., & Abrahamson, D. J. (1988). Trauma and victimization: A model of psychological adaptation. *The Counseling Psychologist*, *16*(4), 531–594. https://doi.org/10.1177/0011000088164002

Mikolajczak, M., & Roskam, I. (2018). A theoretical and clinical framework for parental burnout: The balance between risks and

resources (BR2). *Frontiers in Psychology*, *9*, 886. https://doi.org/10.3389/fpsyg.2018.00886

Mikolajczak, M., Brianda, M. E., Avalosse, H., & Roskam, I. (2018). Consequences of parental burnout: Its specific effect on child neglect and violence. *Child Abuse & Neglect*, *80*, 134–145. https://doi.org/10.1016/j.chiabu.2018.03.025

Nguyen, T. M., Nguyen, A. H., Tran, P. T. T., & Vo, H. T. P. (2023). Self-compassion reduces parental burnout: Mediation roles of parental psychological control and support. *Personal Relationships*, *30*(1), 262–277. https://doi.org/10.1111/pere.12464

Nocentini, A., Fiorentini, G., Di Paola, L., & Menesini, E. (2019). Parents, family characteristics and bullying behavior: A systematic review. *Aggression and Violent Behavior*, *45*, 41–50. https://doi.org/10.1016/j.avb.2018.07.010

O'Brennan, L. M., Bradshaw, C. P., & Sawyer, A. L. (2009). Examining developmental differences in the social-emotional problems among frequent bullies, victims, and bully/victims. *Psychology in the Schools*, *46*(2), 100–115. https://doi.org/10.1002/pits.20357

Olweus, D. (1993). *Bullying at school: What we know and what we can do*. Blackwell Publishing.

Pellegrini, A. D. (2002). Bullying, victimization, and sexual harassment during the transition to middle school. *Educational Psychologist*, *37*(3), 151–163. https://doi.org/10.1207/S15326985EP3703_2

Peng, C., Wang, L. X., Guo, Z., Sun, P., Yao, X., Yuan, M., & Kou, Y. (2024). Bidirectional longitudinal associations between parental psychological control and peer victimization among Chinese adolescents: The mediating role of basic psychological need satisfaction. *Journal of Youth and Adolescence*, *53*(4), 967–981. https://doi.org/10.1007/s10964-023-01910-7

Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, *88*(5), 879–903. https://doi.org/10.1037/0021-9010.88.5.879

Rajendran, K., Kruszewski, E., & Halperin, J. M. (2016). Parenting style influences bullying: A longitudinal study comparing children with and without behavioral problems. *Journal of Child Psychology and Psychiatry*, *57*(2), 188–195. https://doi.org/10.1111/jcpp.12433

Ren, P., Yang, L., Chen, C., & Luo, F. (2023). Parental control and adolescents' bullying victimization: The moderating role of teacher support. *Current Psychology*, *42*(32), 27952–27964. https://doi.org/10.1007/s12144-022-03864-8

Ren, X., Cai, Y., Wang, J., & Chen, O. (2024). A systematic review of parental burnout and related factors among parents. *BMC Public Health*, *24*(1), 376. https://doi.org/10.1186/s12889-024-17829-y

Roskam, I., Raes, M. E., & Mikolajczak, M. (2017). Exhausted parents: Development and preliminary validation of the Parental Burnout Inventory. *Frontiers in Psychology*, *8*, 163. https://doi.org/10.3389/fpsyg.2017.00163

Roskam, I., Aguiar, J., Akgun, E., Arikan, G., Artavia, M., Avalosse, H., Aunola, K., Bader, M., Bahati, C., Barham, E. J., Besson, E., Beyers, W., Boujut, E., Brianda, M. E., Brytek-Matera, A., Carbonneau, N., César, F., Chen, B. B., Dorard, G., & Mikolajczak, M. (2021). Parental burnout around the globe: A 42-country study. *Affective Science*, *2*(1), 58–79. https://doi.org/10.1007/s42761-020-00028-4

Roskam, I., Gallée, L., Aguiar, J., Akgun, E., Arena, A., Arikan, G., Aunola, K., Bader, M., Barham, E. J., Besson, E., Beyers, W., Boujut, E., Brianda, M. E., Brytek-Matera, A., Carbonneau, N., César, F., Chen, B. B., Dorard, G., Dos Santos Elias, L. C., & Mikolajczak, M. (2022). Gender equality and maternal burnout: A 40-country study. *Journal of Cross-Cultural Psychology*, *53*(2), 157–178. https://doi.org/10.1177/00220221211072813

Sciarrino, N. A., Hernandez, T. E., & Davidtz, J. (2018). *Understanding child neglect: Biopsychosocial perspectives*. Springer.

Sebastian, C., Burnett, S., & Blakemore, S. J. (2008). Development of the self-concept during adolescence. *Trends in Cognitive Sciences*, *12*(11), 441–446. https://doi.org/10.1016/j.tics.2008.07.008

Shek, D. T. L. (2008). Perceived parental control and parent–child relational qualities in early adolescents in Hong Kong: Parent gender, child gender and grade differences. *Sex Roles*, *58*(9–10), 666–681. https://doi.org/10.1007/s11199-007-9371-5

Smith, P. K., López-Castro, L., Robinson, S., & Görzig, A. (2019). Consistency of gender differences in bullying in cross-cultural surveys. *Aggression and Violent Behavior*, *45*, 33–40. https://doi.org/10.1016/j.avb.2018.04.006

Song, T., Wang, W., Chen, S., & Li, Y. (2024). Mothers' parental burnout and adolescents' social adaptation and security: The mediating role of parenting style. *Current Psychology*, *43*(27), 23144–23155. https://doi.org/10.1007/s12144-024-06045-x

Swearer, S. M., & Hymel, S. (2015). Understanding the psychology of bullying: Moving toward a social-ecological diathesis–stress model. *American Psychologist*, *70*(4), 344–353. https://doi.org/10.1037/a0038929

Taraban, L., & Shaw, D. S. (2018). Parenting in context: Revisiting Belsky's classic process of parenting model in early childhood. *Developmental Review*, *48*, 55–81. https://doi.org/10.1016/j.dr.2018.03.006

Teuber, Z., Grüter, S., Wang, Q., Bobrowicz, K., Su, Y., & Greiff, S. (2024). The joys and sorrows of parenting: Parental burnout and parental involvement in Chinese and German parents from a demands-resources perspective. *Journal of Cross-Cultural Psychology*, *55*(1), 47–73. https://doi.org/10.1177/00220221231202731

Tippett, N., & Wolke, D. (2014). Socioeconomic status and bullying: A meta-analysis. *American Journal of Public Health*, *104*(6), e48–e59. https://doi.org/10.2105/AJPH.2014.301960

Van Petegem, S., Brenning, K., Baudat, S., Beyers, W., & Zimmer-Gembeck, M. J. (2018). Intimacy development in late adolescence: Longitudinal associations with perceived parental autonomy support and adolescents' self-worth. *Journal of Adolescence*, *65*, 111–122. https://doi.org/10.1016/j.adolescence.2018.03.008

Vansteenkiste, M., & Ryan, R. M. (2013). On psychological growth and vulnerability: Basic psychological need satisfaction and need frustration as a unifying principle. *Journal of Psychotherapy Integration*, *23*(3), 263–280. https://doi.org/10.1037/a0032359

Vigouroux, S. L., & Scola, C. (2018). Differences in parental burnout: Influence of demographic factors and personality of parents and children. *Frontiers in Psychology*, *9*, 887. https://doi.org/10.3389/fpsyg.2018.00887

Wang, Q., Pomerantz, E. M., & Chen, H. (2007). The role of parents' control in early adolescents' psychological functioning: A longitudinal investigation in the United States and China. *Child Development*, *78*(5), 1592–1610. https://doi.org/10.1111/j.1467-8624.2007.01085.x

Wang, W., Chen, S., Wang, S., Shan, G., & Li, Y. (2023). Parental burnout and adolescents' development: Family environment, academic performance, and social adaptation. *International Journal of Environmental Research and Public Health*, *20*(4), 2774. https://doi.org/10.3390/ijerph20042774

Ward, M. A., Clayton, K., Barnes, J., & Theule, J. (2018). The association between peer victimization and attachment security: A meta-analysis. *Canadian Journal of School Psychology*, *33*(3), 193–211. https://doi.org/10.1177/0829573517715737

Wen, Z., Hau, K. T., & Marsh, H. W. (2004). Structural equation model testing: Cutoff criteria for goodness of fit indices and chi-square test. *Acta Psychologica Sinica*, *36*(2), 186–194.

Wen, W., Sim, L., Hou, Y., Chen, S., & Kim, S. Y. (2023). Change patterns of mother-adolescent perceived parenting and the corresponding trajectories in their internalizing symptoms.

Developmental Psychology, *59*(10), 1906–1920. https://doi.org/10.1037/dev0001620

Woine, A., Dardier, V., Mikolajczak, M., & Roskam, I. (2026). Bidirectional and transactional effects between parental burnout and child internalizing and externalizing behaviors. *Journal of Family Psychology*, *40*(3), 447–457. https://doi.org/10.1037/fam0001414

Wu, W., Guo, Z., Li, S., Tu, F., Wu, X., Ma, X., Teng, Z., Chen, Y., & Zeng, Y. (2022). The influence of parental autonomy support on cyberbullying victimization of high school students: A latent moderation analysis. *Acta Psychologica*, *230*, 103739. https://doi.org/10.1016/j.actpsy.2022.103739

Yan, T., Hou, Y., & Deng, M. (2026). The effect of parental burnout on psychological adjustment among Chinese children with developmental disabilities: The roles of parental autonomy support and psychological control. *International Journal of Developmental Disabilities*, *72*(2), 251–264. https://doi.org/10.1080/20473869.2024.2331830

Yang, B., Chen, B. B., Qu, Y., & Zhu, Y. (2021). Impacts of parental burnout on Chinese youth's mental health: The role of parents' autonomy support and emotion regulation. *Journal of Youth and Adolescence*, *50*(8), 1679–1692. https://doi.org/10.1007/s10964-021-01450-y

Yang, P., Schlomer, G. L., & Lippold, M. A. (2023). Mothering versus fathering? Positive parenting versus negative parenting? Their relative importance in predicting adolescent aggressive behavior: A longitudinal comparison. *Developmental Psychology*, *59*(1), 69–83. https://doi.org/10.1037/dev0001442

Yang, Z., Luo, Y., Chen, F., Qi, M., Luo, R., Li, Y., & Wang, Y. (2024). Environmental unpredictability and self-concept clarity in Chinese adolescents during the COVID-19 pandemic: The mediating role of depressive symptoms. *Current Psychology*, *43*(29), 24159–24167. https://doi.org/10.1007/s12144-024-06057-7

Zhang, W., Wu, J., & Jones, K. (1999). A revision of the Chinese version of the Olweus Child Bully/Victim Questionnaire. *Psychological Development and Education*, *15*(2), 7–11. https://doi.org/10.16187/j.cnki.issn1001-4918.1999.02.002

Zhang, S., Hong, J. S., Garthe, R. C., Espelage, D. L., & Schacter, H. L. (2021). Parental stress and adolescent bullying perpetration and victimization: The mediating role of adolescent anxiety and family resilience. *Journal of Affective Disorders*, *290*, 284–291. https://doi.org/10.1016/j.jad.2021.04.023

Zhang, H., Wang, W., Liu, S., Feng, Y., & Wei, Q. (2023). A meta-analytic review of the impact of child maltreatment on self-esteem: 1981 to 2021. *Trauma Violence & Abuse*, *24*(5), 3398–3411. https://doi.org/10.1177/15248380221129587

Zhang, R., Xie, R., Feng, J., Ding, W., Song, S., Yang, Q., & Li, W. (2025). The bidirectional relationship between parental psychological control, negative automatic thoughts, and bullying in Chinese children: A 2-year longitudinal study. *Psychology of Violence*, *15*(5), 568–577. https://doi.org/10.1037/vio0000585

Zheng, J., & Chen, B. B. (2025). Parent-adolescent discrepancies in perceiving parental psychological control and autonomy support predict adolescents' psychological adjustment: Does adolescent gender make a difference? *Journal of Youth and Adolescence*, *54*(6), 1588–1599. https://doi.org/10.1007/s10964-025-02144-5

Zhou, P., Dong, J., Liu, J., Wen, H., & Wang, Z. (2023). The relationship between parent–child relationship and peer victimization: A multiple mediation model through peer relationship and depression. *Frontiers in Behavioral Neuroscience*, *17*, 1170891. https://doi.org/10.3389/fnbeh.2023.1170891

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Haotian Bian is a PhD student in the Collaborative Innovation Center of Assessment for Basic Education Quality at Beijing Normal University. His current research focuses on parenting, family dynamics and adolescent mental health.

Qian Pu is a Master's student in the Collaborative Innovation Center of Assessment for Basic Education Quality at Beijing Normal University. Her major research interests include family relationships, personal values and adolescent mental health.

Ying He is a Master's student in the Collaborative Innovation Center of Assessment for Basic Education Quality at Beijing Normal University. Her major research interests include personal values and family context influences on adolescents' mental health and developmental outcomes.

Yifan Zhang is a PhD student in the State Key Laboratory of Cognitive Neuroscience and Learning at Beijing Normal University. Her major research interests include sibling relationships, parental educational anxiety and adolescent mental health.

Fumei Chen is an associate professor in the Collaborative Innovation Center of Assessment for Basic Education Quality at Beijing Normal University. Her major research interests include family and the development of children and adolescents.